MILITARY SPECIFICATION

LAUNCHERS, ROCKET

LAU-10C/A AND LAU-10D/A

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE.

1.1 Scope. This specification covers two dual purpose airborne external store rocket launchers capable of shipping and storing four 5.0-inch ZUNI Rocket Motors, and launching four 5.0-inch ZUNI rockets.

2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-5015	Connector, Electric, AN Type.
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles.
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment, Including Ground Support Equipment.
MIL-P-19834	Plates, Identification, Metal Foil, Adhesive Backed, Military Specification for.

FSC 1055

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Naval Air Systems Command
(Code Ident 30003)

AS-1142	Critical Item Product Function Specification for Intervalometer, Rocket Launcher.

STANDARDS

Federal

FED-STD-151	Metals, Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-1648	Criteria and Test Procedure for Ordnance Exposed to an Aircraft Fuel Fire.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

DL 4902046	Launcher, Rocket LAU-10C/A.
DL 4902292	Launcher, Rocket LAU-10D/A.

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OTHER PUBLICATIONS

Naval Air Systems Command
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NAVAIR 11-75A-63	5.0" Airborne Rocket Launcher LAU-10/33/35 Series.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specified procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 Preproduction sample. Unless otherwise specified in the contract or purchase order, a preproduction sample consisting of four launchers shall be delivered to the testing activity designated in the contract or purchase order. The preproduction sample shall be manufactured using the same methods, materials, and procedures proposed for production. Any production prior to acceptance of the preproduction sample is at the risk of the contractor.

3.2 Conformance to drawings. The launcher shall be fabricated and assembled in accordance with the drawings and other documents listed on DL 4902046 for LAU-10C/A and DL 4902292 for LAU-10D/A.

3.3 Materials and processses.

3.3.1 Materials. All materials used in the construction of the launcher shall be in accordance with MIL-STD-454, Requirement 3, Flammable Materials; Requirement 4, Fungus-Inert Materials; Requirement 16, Dissimilar Metals.

3.3.2 Processes. Unless otherwise approved in writing by the procuring activity, the manufacturing processes and finishes shall be as specified herein, and on the applicable drawing.

3.4 Design and construction.

3.4.1 Interchangeability. All parts having the same part number shall be directly and completely interchangeable with each other in respect to fit, function, and performance. All applicable provisions of MIL-I-8500 shall apply.

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3.4.2 Nomenclature and nameplates. Nomenclature shall be in accordance with MIL-N-18307, and nameplates shall be in accordance with MIL-P-19834.

3.4.3 Reliability. The reliability requirement for the launcher, less intervalometer, shall be 0.9904 at a 90 percent confidence level.

3.4.4 Weight. The weight of the launcher including shock pans, but without rockets, shall be as follows:

Configuration	Weight (pounds)
LAU-10C/A	<141
LAU-10D/A	<175

3.4.5 Electrical connectors. All electrical connectors shall conform to MIL-C-5015.

3.5 Performance.

3.5.1 Firing circuits. The maximum allowable wiring circuit continuity resistance shall be 0.05 ohm. During firing, whether in single or ripple fire mode, the open circuit voltage from any output circuit to ground, other than that circuit which is energized, shall not exceed 0.005 volt (V) positive or negative. This includes alternating current (ac), direct current (dc) or random voltage. In addition, the requirements of 3.5.3.2 shall be met.

3.5.2 Detents. The detent shall not release with a pull of less than 1865 pounds total load, and shall not fail to release with a pull of 2400 pounds total load.

3.5.3 Functional.

3.5.3.1 Loading. The launcher shall be capable of being loaded and unloaded as described in NAVAIR 11-75A-63.

3.5.3.2 Electrical. The electrical tests of the launcher shall be conducted with the intervalometer installed. The intervalometer shall be an item that has met the requirements of AS-1142.

3.5.3.2.1 Single fire. During single fire, when the launcher is energized, the output pulse for the first firing circuit shall appear within 20 milliseconds (ms) after power is applied. The pulse duration

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shall be a minimum of 20 ms. When the input power is removed and reapplied, the next firing circuit shall be energized. The output pulse shall appear on this circuit within 20 ms after power is applied, for a minimum duration of 20 ms. The intervalometer shall continue to put out a 20 ms pulse (within 20 ms) to the firing circuits (in sequence) with each application of power. The output pulse shall be not less than 1.5 volts direct current (Vdc) when measured across the specified load.

3.5.3.2.2 Ripple fire. During ripple fire, when the launcher is energized, the output pulse for the first firing circuit shall appear within 20 ms after power is applied. The pulse duration shall be a minimum of 20 ms. The intervalometer shall continue the cycle, applying an output pulse to each of the firing circuits until the load position is reached, at which time it will stop. After the intervalometer has stopped at the load position, and power is removed and reapplied, the intervalometer shall again cycle through the four firing circuits. The time interval between the leading edge of any given pulse and the leading edge of the next pulse shall be 95 ± 15 ms. The output pulse shall be not less than 1.5 Vdc when measured across the specified load.

3.5.4 Environmental.

3.5.4.1 Leakage. The 8-hour air leakage of the launcher, with shock pans installed and internal air pressure reduced by 7.0 ± 0.5 inches of mercury, shall not exceed 4.5 inches of mercury.

3.5.4.2 Palletized transportation vibration. There shall be no scores in the rear 10 inches of any launcher tube more than 0.020-inch deep. During and after the palletized transportation test, all rockets shall remain detented, and the launcher shall meet the requirements of 3.5.3.1. The LAU-10D/A launcher coating shall remain intact.

3.5.4.3 Palletized rough handling. Failure of the shipping pallet to hold undeformed shock pans in place will not be considered a launcher failure. There shall be no obvious cracks in the launcher end castings. During and after the palletized rough handling test, the shock pans shall not deform sufficiently to allow any launcher to move out of position. All rockets shall remain detented, and the launcher shall meet the requirements of 3.5.3.1. The LAU-10D/A launcher coating shall remain intact.

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3.5.4.4 Humidity in salt-laden atmosphere. After the humidity in salt-laden atmosphere test, the launcher shall meet the requirements of 3.5.1, 3.5.2 and 3.5.3.1. The LAU-10D/A launcher coating shall remain intact.

3.5.4.5 Catapult launch and arrested landing. During and after the test, there shall be no obvious cracks in the launcher end castings. All rockets shall remain detented during test, and the launcher shall meet the requirements of 3.5.2 and 3.5.3.1. The LAU-10D/A launcher coating shall remain intact.

3.5.4.6 Aircraft and helicopter vibration. During and after the vibration test there shall be no scores in the rear 10 inches of any launcher tube more than 0.020-inch deep. All rockets shall remain detented during test, and the launcher shall meet the requirements of 3.5.1, 3.5.2, 3.5.3.1, and 3.5.3.2. The LAU-10D/A launcher coating shall remain intact.

3.5.5 Static loads.

3.5.5.1 Launcher. After the static loads test, the launcher shall show no evidence of permanent deformation and shall meet the requirements of 3.5.2 and 3.5.3.1.

3.5.5.2 Fairings. After the static loads test, the fairings shall show no evidence of permanent deformation other than a slight pull of fairing material at the fairing band rivets.

3.5.6 Operational.

3.5.6.1 Ground firing. The launcher shall fire all the rockets in both single fire and ripple fire modes, and meet the requirements of 3.5.3.2. During and after the ground firing test, the launcher (and coating on the LAU-10D/A) shall remain intact with no obvious cracks in the launcher end castings. The launcher shall meet the requirements of 3.5.1 and 3.5.3.2.

3.5.6.2 Aircraft firing. The launcher shall fire out properly and meet the requirements of 3.5.3.2. During and after the aircraft firing test, the launcher (and coating on the LAU-10D/A) shall remain intact with no obvious cracks in the launcher end castings. The launcher shall meet the requirements of 3.5.1 and 3.5.3.2.

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3.5.6.3 Fast cookoff (LAU-10D/A only). The fast cookoff test shall meet the requirements of MIL-STD-1648.

3.6 Workmanship. The workmanship shall meet the requirements of MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to observe and to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of examination and tests. The inspections of the launcher shall be classified as follows:

(a) Preproduction inspections (see 4.3).

(b) Quality conformance inspections (see 4.4).

(1) Individual inspections (see 4.4.1).

(2) Acceptance inspections (see 4.4.2).

4.3 Preproduction inspections. Unless otherwise specified in the contract or purchase order, the preproduction sample shall be subjected to the inspections of Table I, in the sequence shown, at a facility designated in the contract or purchase order.

4.4 Quality conformance inspections. The contractor shall be responsible for accomplishing individual acceptance inspections. The launchers shall be submitted for acceptance by lots, and the launchers comprising each lot shall be those having passed the individual inspections. Acceptance of each lot will be determined by the launchers successfully passing the acceptance inspection of 4.4.2. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

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Table I. Preproduction and Quality Conformance Inspections

Examination or test	Requirement paragraph	Method paragraph
Individual inspections		
Visual examination	3.2	4.5.1
Certification	3.3	4.5.2
Workmanship	3.6	4.5.1
Preparation for delivery	Section 5	4.9
Firing circuits	3.5.1	4.5.3
Detents	3.5.2	4.5.4
Functional tests		
Loading	3.5.3.1	4.5.5
Electrical	3.5.3.2	4.5.6
Environmental tests		
Leakage	3.5.4.1	4.6.2
Palletized transportation vibration	3.5.4.2	4.6.1.1
Palletized rough handling	3.5.4.3	4.6.1.2
Leakage	3.5.4.1	4.6.2
Detents	3.5.2	4.5.4
Loading	3.5.3.1	4.5.5
Humidity in salt-laden atmosphere	3.5.4.4	4.6.3
Detents	3.5.2	4.5.4
Loading	3.5.3.1	4.5.5
Catapult launch and arrested landing	3.5.4.5	4.6.1.3
Detents	3.5.2	4.5.4
Loading	3.5.3.1	4.5.5
Aircraft and helicopter vibration	3.5.4.6	4.6.1.4
Detents	3.5.2	4.5.4
Loading	3.5.3.1	4.5.5
Electrical	3.5.3.2	4.5.6
Static loads	3.5.5	4.6.4
Detents	3.5.2	4.5.4
Loading	3.5.3.1	4.5.5
Operational tests		
Ground firing	3.5.6.1	4.7.1
Aircraft firing	3.5.6.2	4.7.2
Fast cookoff	3.5.6.3	4.7.3
Reliability	3.4.3	4.7.4

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4.4.1 Individual inspections. Individual inspections shall consist of the examinations and tests specified in 4.5 and are to be performed on each launcher.

4.4.2 Acceptance inspection. The lot shall be randomly sampled in accordance with 4.4.2.1, and shall be subjected to all the inspections of 4.5, 4.6, and 4.7 to determine that production launchers continue to be equivalent in performance and construction to the approved preproduction sample.

4.4.2.1 Sampling. Sampling for acceptance inspections shall be in accordance with MIL-STD-105, Normal Inspection Level II, for an Acceptable Quality level (AQL) of 2.5 percent defective.

4.4.2.2 Lot. The lot size and formation shall be as defined in MIL-STD-105.

4.5 Individual inspections.

4.5.1 Inspections. Prior to assembly, inspections shall be made to ascertain that all parts and components conform to the specifications and drawings listed on DL 4902046 for LAU-10C/A, and DL 4902292 for LAU-10D/A and as specified in 3.6.

4.5.2 Certification. Unless otherwise specified in the contract or purchase order, the contractor shall have written certification available for the Government. This written certification shall be accompanied by objective quality evidence (as defined in MIL-STD-109) that the materials, parts, and processes conform to the prescribed requirements.

4.5.3 Firing circuits. The wiring of the circuits shall be checked for continuity. Each of the firing circuits shall be checked between the intervalometer connector and its respective firing contact. The ground circuit shall be checked between the intervalometer connector and each of the detent pawls. The launcher shall meet the requirements of 3.5.1, as applicable.

4.5.4 Detents. Place the launcher in a test fixture, approved by the procuring activity, and exert 1865 ± 10 pounds pull then 2400 ± 10 pounds pull on each detent. To be acceptable, the launcher shall meet the requirements of 3.5.2.

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4.5.5 Loading. The launcher shall be loaded with four ZUNI rockets and then unloaded as described in NAVAIR 11-75A-63. The launcher shall meet the requirements of 3.5.3.1.

4.5.6 Electrical. The launcher shall meet the requirements of 3.5.3.2.

(a) When energized by a power source having the following characteristics:

(1) Voltage range of 26 ± 3 Vdc.

(2) A 5.0 ± 0.5 ohms, noninductive resistance (50 watts minimum) connected, in the positive lead, in series with the intervalometer. All input voltage measurements shall be made from the junction of the power source and the 5-ohm resistor measured to negative ground.

(3) A short circuit protection with a response time of 100 ms minimum and a cutoff current of not less than 10 amperes.

(b) When each of the firing circuits are connected to a load consisting of either a 0.5 ± 0.01 ohm noninductive resistance applied for not less than 10 ms during the time the output pulse is applied to the firing circuit, or a rocket igniter.

4.6 Environmental tests.

4.6.1 During the following tests, load the launcher with four all-up inert rockets weighted to simulate live-loaded weight. The rockets consist of MK 71 MOD 1 motors, MK 63 warheads, and MK 93 fuzes, all inert.

4.6.1.1 Palletized transportation vibration. Subject four inertly loaded and palletized launchers (one pallet load) to 45 minutes of sinusoidal vibration along each of the mutually perpendicular axes in accordance with MIL-STD-810, Method 514, Weight Limited, Resonance Dwell Eliminated. Excitation time shall be equally divided between temperature of - 40 ± 2 degrees centigrade (°C) (- 40 ± degrees Fahrenheit (°F)) an ambient (23 ± 10°C or 73 ± 18°F). Excitation inputs levels shall be 0.65 gravity unit (g) peak amplitude from 5 to

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27 Hertz (Hz) and 0.018-inch double amplitude from 27 to 52 Hz. Sweep time from 5 to 26 to 5 Hz shall be 15 minutes logarithmically. The launcher shall meet the requirements of 3.5.4.2 (see Note 4.6.1.2).

4.6.1.2 Palletized rough handling. Subject four inertly loaded launchers (one pallet load) to a total of six impacts in different locations from a height of 18 inches in accordance with MIL-STD-810, Method 516, Procedure II. All impacts shall be on the base of the pallet. Impacts shall be as follows:

(a) Two edges.

(b) Two ends.

(c) Two diagonally opposing corners.

Dimensions and materials of the impact surface shall be approved by the procuring activity prior to testing. The launcher shall meet the requirements of 3.5.4.3.

NOTE. Perform the tests of 4.6.1.1 and 4.6.1.2. Then, perform the second leakage test, 4.6.2, and verify that the requirements of 3.5.4.2 and 3.5.4.3 are met.

4.6.1.3 Catapult launch and arrested landing. Attach the inertly loaded launcher to an existing or simulated and approved bomb rack. Subject the loaded launcher to a total of 18 half-sine wave shock pulses at ambient temperature (23 ± 10°C or 73 ± 18°F). Three shocks shall be applied along each direction of each of three mutually perpendicular axes. Shock pulse levels shall be as follows:

(a) Longitudinal: 13.5 + 2.2, - 0.0 g, 25 ms.

(b) Vertical: 13.8 + 2.7, - 0.0 g, 25 ms.

(c) Lateral: 8.6 + 1.3, - 0.0 g, 25 ms.

The launcher shall meet the requirements of 3.5.4.5.

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4.6.1.4 Aircraft and helicopter vibration. The launcher shall be subjected to the following two-part vibration test. The launcher shall meet the requirements of 3.5.4.6.

4.6.1.4.1 High speed aircraft vibration. The launcher shall have random excitation applied for 45 minutes in each axis at each of two temperature extremes, - 40 ± 2°C (- 40 ± 4°F) and 60 ± 2°C (140 ± 4°F). In the transverse and verticalaxes, excitation input shall be 0.015 g^2/Hz power spectral density flat from 200 to 2,000 Hz with 3 decibels in each octave (dB/octave) rolloff from 10 to 200 Hz. In the longitudinal axis, excitation input shall be 0.010 g^2/Hz power spectral density flat from 200 to 2,000 Hz with a 3 dB/octave rolloff from 10 to 200 Hz.

4.6.1.4.2 Helicopter. The launcher shall have sinusoidal excitation applied at 25-minute frequency dwells as follows:

Vertical axis		Transverse axis		Longitudinal axis	
*PA	Hz	PA	Hz	PA	Hz
1.8 g	11	0.7 g	11	1.0 g	11
0.9 g	22	0.6 g	22	0.6 g	22
				0.6 g	44

*Peak amplitude

Apply 5-minute frequency sweeps covering the frequency range of 5 to 500 to 5 Hz in 5 minutes along each of three mutually perpendicular axes with the launcher first fully loaded with four inert all-up rounds, then with the launcher empty. Total test time for this portion of the test shall be 10 minutes per axis. Excitation inputs shall be as follows:

Vertical axis: 0.40-inch double amplitude from 5 to 10 Hz, 2 g peak amplitude from 10 to 500 Hz.

Transverse axis: 0.40-inch double amplitude from 5 to 7 Hz, 10 g peak amplitude from 7 to 18 Hz, 0.06-inch double amplitude from 18 to 26 Hz, and 2 g peak amplitude from 26 to 500 Hz.

Longitudinal axis: 0.40-inch double amplitude from 5 to 7 Hz, 1.0 g peak amplitude from 7 to 33 Hz, 0.018-inch double amplitude from 33 to 48 Hz, and 2 g peak amplitude from 48 to 500 Hz.

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4.6.2 Leakage. The empty launcher, with shock pans installed, shall be temperature conditioned at ambient temperature (23 ± 10°C or 73 ± 18°F) for 24 hours. Evacuate the launcher to a vacuum of 7 ± 0.5 inches of mercury, and continue to store the launcher at the conditioning temperature for a period of 8 hours. The launcher shall meet th requirements of 3.5.4.1.

4.6.3 Humidity in salt-laden atmosphere. The empty launcher, shock pans, thermal shields, and fairings removed, shall be subjected to ten 12-hour cycles of humidity with salt-laden atmosphere. Prior to beginning the first test cycle, precondition the chamber at 21 ± 1°C (70 ± 2°F). Each cycle shall be as follows:

(a) Spray a synthetic sea water solution, prepared in accordance with FED-STD-151, Method 812.1, into the test chamber and within 30 minutes stabilize at 21 ± 1°C (70 ± 2°F) at 100 percent relative humidity (rh). Maintain the condition for a total, including stabilization time, of 5 hours.

(b) Change the temperature to 35 ± 1°C (95 ± 2°F) at 95 percent rh.

(c) Discontinue spray and change the temperature to 57 ± 1°C (135 ± 2°F) at 30 percent rh within 30 minutes. Maintain this condition for 5 hours after stabilization.

(d) Change the temperature to 21 ± 1°C (70 ± 2°F) at 100 percent rh within 1 hour.

Total test time shall be 120 hours. The launcher shall meet the requirements of 3.5.4.4.

4.6.4 Static loads.

4.6.4.1 Launcher. The empty launcher shall be suspended from a bomb rack or simulated bomb rack with a 14-inch suspension. The test loads shall be applied in such a manner that the loads are equally divided along the four launching tubes and uniformly distributed over the length of the tubes. Three load forces shall be applied simultaneously at a point as near as possible to the center of gravity of the total mass. The loads applied and direction shall be as follows:

(a) 9,450 ± 30 pounds downward.

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(b) 6,000 ± 30 pounds to the side in each direction (left to right and right to left).

(c) 7,200 ± 30 pounds forward.

(d) 7,200 ± 30 pounds rearward.

The test shall be in two parts as follows:

(e) Down load and side load, right to left, combined with forward load.

(f) Down load and side load, left to right, combined with rearward load.

Repeat above test using 30-inch suspension. The launcher shall meet the requirements of 3.5.5.1.

4.6.4.2 Fairings. The fairings, installed on a launcher or a simulated launcher end, shall be subjected to static loads simulating aerodynamic loads as follows:

(a) Axial (externally applied): 1,390 pounds total load applied in such a manner that the peak load force is at the tip of the cone and gradually diminishes to zero at the base, with equal forces applied over the periphery of the cone.

(b) Side (externally applied): Side load of 437 pounds total load equally distributed from the tip of the cone to the base shall be applied in a manner that also distributes the load over approximately one-half of the circumference.

The fairings shall meet the requirements of 3.5.5.2.

4.7 Operational tests.

4.7.1 Ground firing. The launcher with fairings shall be loaded with four all-up rockets consisting of live MK 71 MOD 1 motors, and inert warheads and fuzes. The launcher shall be fired from a suitable ground firing mount. All four rounds shall be fired in the single fire mode. The launcher shall then be reloaded and four rounds fired in the ripple fire mode. The launcher shall meet the requirements of 3.5.6.1.

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4.7.2 Aircraft firing. The launcher with fairings shall be loaded with four all-up rockets consisting of live MK 71 MOD 1 motors, and inert warheads and fuzes. The launcher shall be attached to the normal carry stations on a military jet attack or fighter aircraft (6.2(d)) and fired using simulated combat flight profiles. All four rounds shall be fired in single fire mode. The launcher shall be reloaded and fired in the ripple fire mode. The launchers shall meet the requirements of 3.5.6.2.

4.7.3 Fast cookoff. The LAU-10D/A launcher, loaded with four live all-up rockets consisting of MK 71 MOD 1 motor, MK 63 warhead, and MK 93 fuze, and with thermally protected front fairing and with rear thermal shield, shall be subjected to the conditions of MIL-STD-1648.

4.7.4 Reliability demonstration. The data from testing performed in Table I shall be used to demonstrate the reliability requirement of 3.4.3.

4.8 Test plan and test reports.

4.8.1 Test plan. Unless otherwise specified in the contract or purchase order, a detailed test plan shall be provided to the procuring activity prior to testing and delivery of the preproduction sample.

4.8.2 Test report. When specified in the contract or purchase order, test reports in accordance with MIL-STD-831 shall be available at the contractor's plant to the procuring activity. When required by the contract or purchase order, copies shall be delivered to the procuring activity.

4.9 Preparation for delivery. Prior to shipment, the launcher shall be inspected to ensure that preservation, packaging, packing, and marking for shipment are in accordance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY.

5.1 Palletizing. The complete launcher shall be palletized for delivery in accordance with procuring activity instructions on Government-furnished pallets. Delivery shall be as directed by the procuring activity.

5.2 Marking and labeling.

5.2.1 Pallets. Marking and labeling shall be in accordance with MIL-STD-129.

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6. NOTES.

6.1 Intended use. These launchers are intended for shipping and storing four 5.0-inch ZUNI rocket motors, and for launching four 5.0-inch ZUNI rockets.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Preproduction sample testing activity (see 3.1 and 4.3).

(c) Certification of conformance (see 4.5.2).

(d) Designated activity or activities and aircraft for firing tests (see 4.7).

(e) Test reports, requirement for and distribution of (see 4.8.2).

(f) Any preparation for delivery requirement other than Section 5.

(g) A quality program in accordance with ASPR 14-101.4, after coordination in accordance with ASPR 14-101.5(a)i(D).

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DOCUMENT IDENTIFIER AND TITLE
MIL-L-81979(AS) LAUNCHERS, ROCKET LAU-10C/A and LAU-10D/A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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